Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- **Foundational Concepts of Systems of Linear Equations**
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • 16 min

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Foundational Concepts of Systems of Linear Equations

0:06 / 0:58 2x

Read Section 2.1: “Systems of Linear Equations” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this section, you will explore the foundational concepts of systems of linear equations, which are essential for solving many practical problems in machine learning and data analysis. You will learn how to represent and solve these systems and understand their significance in the context of linear algebra.

As you read, consider the following questions:

- How do systems of linear equations help in solving real-world problems?
- What are the different methods for solving systems of linear equations, and how do they compare in terms of efficiency and accuracy?
- What are the implications of having no solution, a unique solution, or an infinite number of solutions in a system of linear equations?
- How can we interpret the geometric representation of systems of linear equations?

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